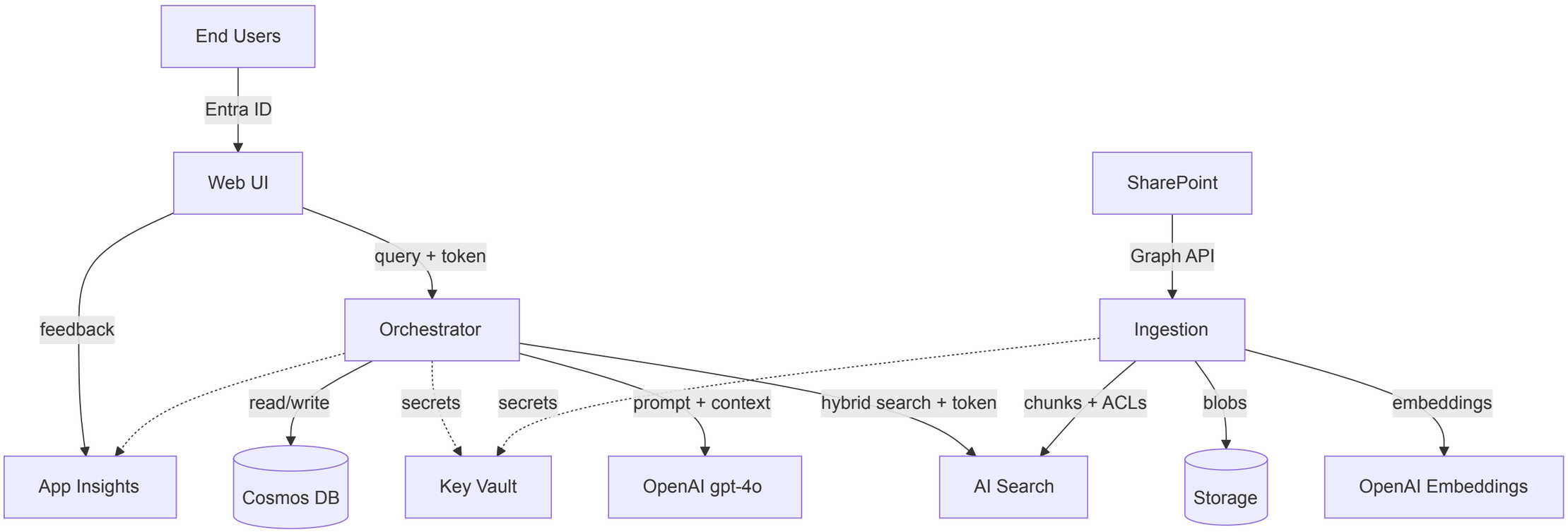
GPT-RAG accelerator

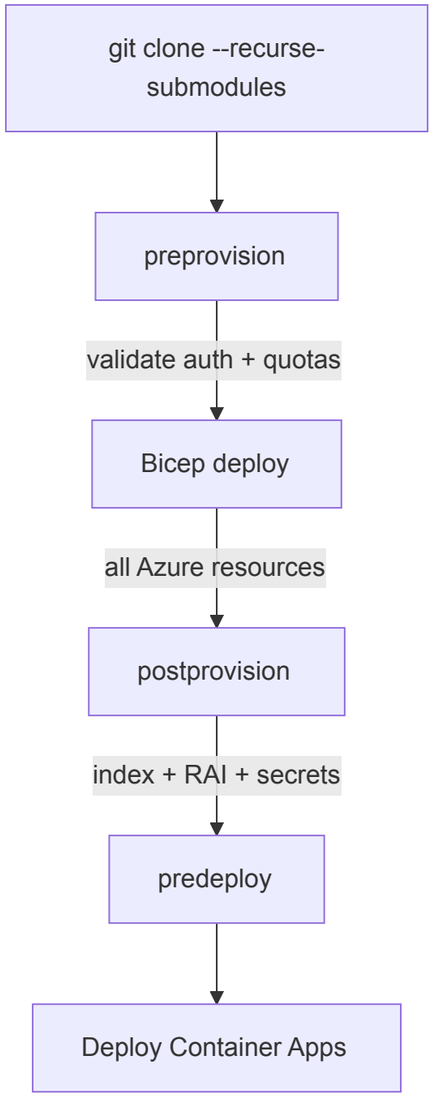
GPT-RAG Solution Accelerator — Architecture Overview

> **Source:** github.com/Azure/gpt-rag (v2.5.0)
> **Strategy:** Single Agent + SharePoint only (initial stage)

High-Level Architecture



Deployment Flow with azd



What each hook does

Hook	Actions
preprovision	Validate Azure CLI auth, check quotas, verify prerequisites
Bicep	Deploy Hub + Project, OpenAI, AI Search, Cosmos DB, Key Vault, Container Apps Env, VNet, Managed Identities
postprovision	Create search index from `search.j2`, RAI setup via `setup.py`, Key Vault secrets, Knowledge Base/Source
predeploy	Dependency checks
Deploy	Push orchestrator, ingestion, frontend as Container Apps

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Configuration Points

Files You Edit (in the repo)

File	Path	What You Configure
`search.j2`	`config/search/`	Index schema, vector dims, analyzer, security fields, semantic config
`search.settings.j2`	`config/search/`	`ENABLE_AGENTIC_RETRIEVAL` , embedding dimensions, toggles
`raiblocklist.json`	`config/aifoundry/`	Organization-specific blocked terms
`raipolicies.json`	`config/aifoundry/`	Severity thresholds: hate, sexual, violence, self-harm
`azure.yaml`	Root	Service definitions, hooks (rarely changed)

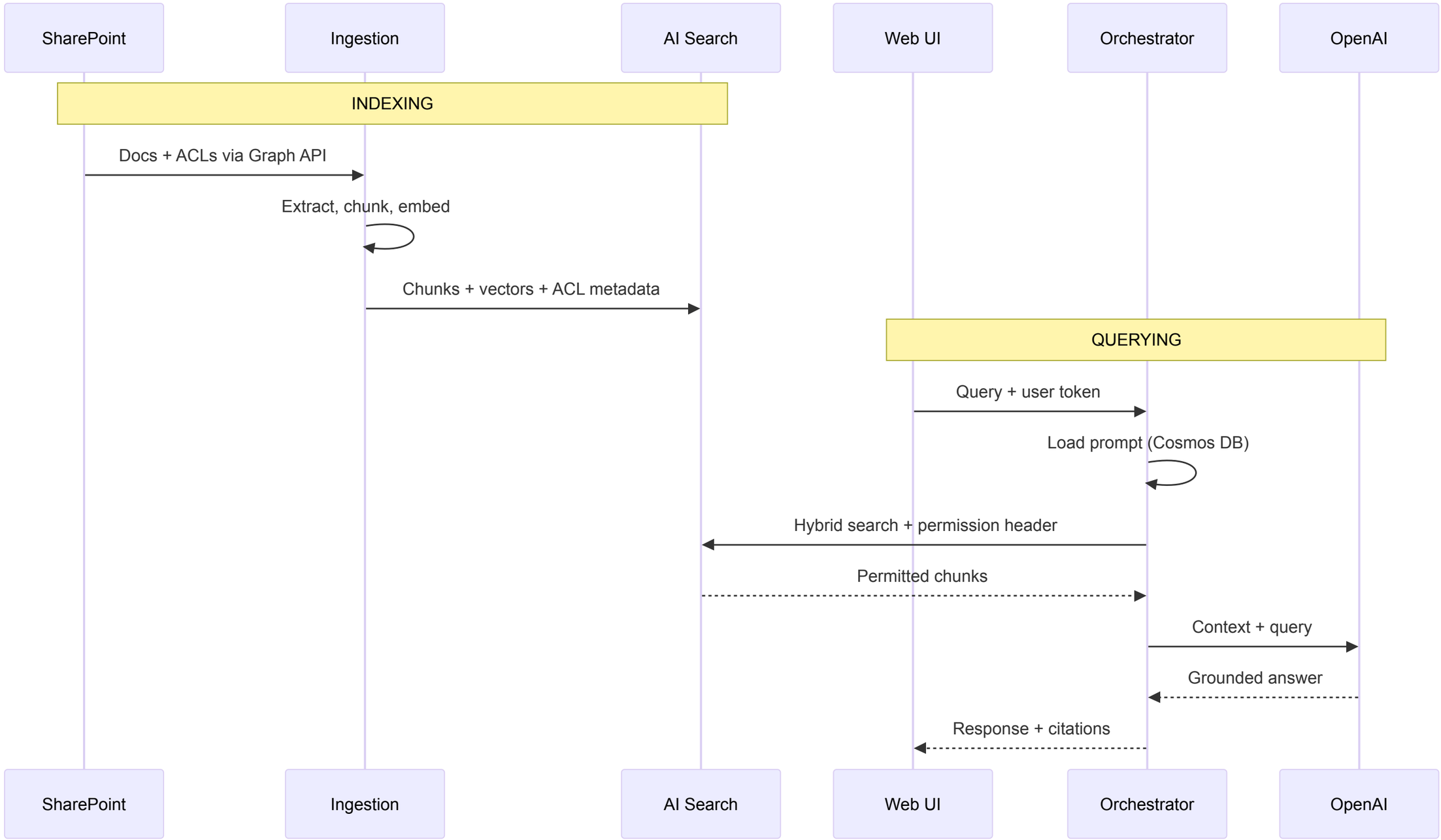
Azure Portal / CLI Settings

Setting	Where	What You Configure
OpenAI Quota	Azure Portal	TPM for gpt-4o + embedding model, region
Entra ID	Azure Portal	App Registration, API Permissions (Sites.Read.All, Files.Read.All)
AI Search Tier	azd env / Bicep	Basic (dev), S1 (small prod), S2 (medium prod)
azd env variables	CLI	`AZURE_LOCATION` , model names, feature flags

Organizational Decisions

Decision	Impacts	Options
SharePoint scope	Ingestion + Entra permissions	Which sites, which libraries, include Lists?
Document security	Index schema + identity propagation	All users see all docs vs permission-aware filtering
Embedding model	search.j2 dims + cost + quality	large 3072d (recommended) vs small 1536d
RAI thresholds	raipolicies.json + raiblocklist.json	Content filter levels + blocked terms
Azure region	All resources	Must support gpt-4o + embeddings + AI Search
Network mode	Bicep landing zone	Private endpoints (prod) vs public (dev)

Data Flow: Indexing and Querying



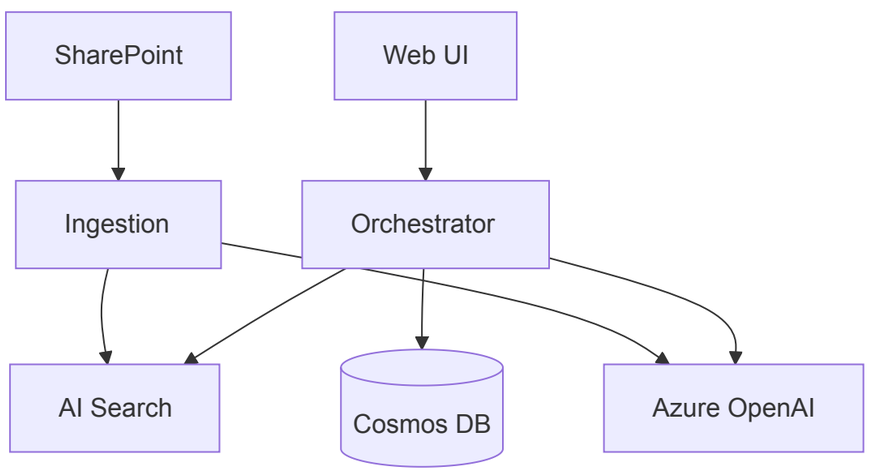
Repository Structure

```
github.com/Azure/gpt-rag (v2.5.0)
│
├── azure.yaml                    ← azd project definition + hooks
├── manifest.json                 ← component versions
│
├── config/
│   ├── search/
│   │   ├── search.j2             ← INDEX SCHEMA (most important config)
│   │   └── search.settings.j2    ← feature toggles
```

```
│   └── aifoundry/
│       ├── setup.py              ← RAI automation script
│       ├── raiblocklist.json     ← blocked terms
│       └── raipolicies.json      ← content filter thresholds
│
├── infra/
│   └── (bicep-ptn-aiml-landing-zone)  ← Git submodule, all Bicep here
│
├── scripts/
│   ├── preprovision.sh           ← pre-deploy validation
│   ├── postprovision.sh          ← index + RAI + secrets setup
│   └── predeploy.sh             ← dependency checks
│
└── components/
    ├── gpt-rag-orchestrator/     ← Agent logic (Python)
    ├── gpt-rag-ingestion/        ← Doc processing (Python)
    └── gpt-rag-ui/               ← Chat frontend (Python)
```

Your Initial Scope

Active now — what gets deployed and used from day one:



You configure:

What	File / Setting
Index schema	`search.j2`
RAI filters	`raiblocklist.json`, `raipolicies.json`
Entra ID	App registration + API permissions
SharePoint	Which sites and libraries to index
System prompt	Cosmos DB `prompts` container
azd env	Region, model names, feature flags

Available later (not deployed in initial scope):

Component	When to add
gpt-rag-mcp	When you need external API/tool integration
NL2SQL Indexes	When you want natural language database queries
Agentic Retrieval	When you want AI Search to plan sub-queries autonomously

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Cosmos DB: Prompts and Runtime Data

GPT-RAG provisions a Cosmos DB account automatically during `azd provision`. It contains four containers:

Container	Purpose	Who manages it
`prompts`	**System prompts** — persona, tone, citation rules, guardrails	Content owner / project lead
`conversations`	Chat history per user session	Automatic (orchestrator writes)
`datasources`	Data source configuration	Admin during setup
`mcp`	MCP tool definitions (not used in Single Agent)	Developer (later)

Custom Prompts (no code needed)

The `prompts` container is where you define **how the chatbot behaves**: its persona, language, citation style, and domain-specific guardrails. You manage these via the **GPT-RAG Web UI** or directly in the **Azure Portal Cosmos DB Data Explorer**.

Changes take effect at runtime — no redeployment needed.

What to configure in prompts:

- System persona ("You are a helpful assistant for [organization]...")
- Citation behavior (how strictly to cite sources, what to do with no results)
- Tone and language (formal vs conversational, multilingual support)
- Domain guardrails ("only answer questions about X", "never provide legal advice")
- Response format preferences

Preparation Checklist

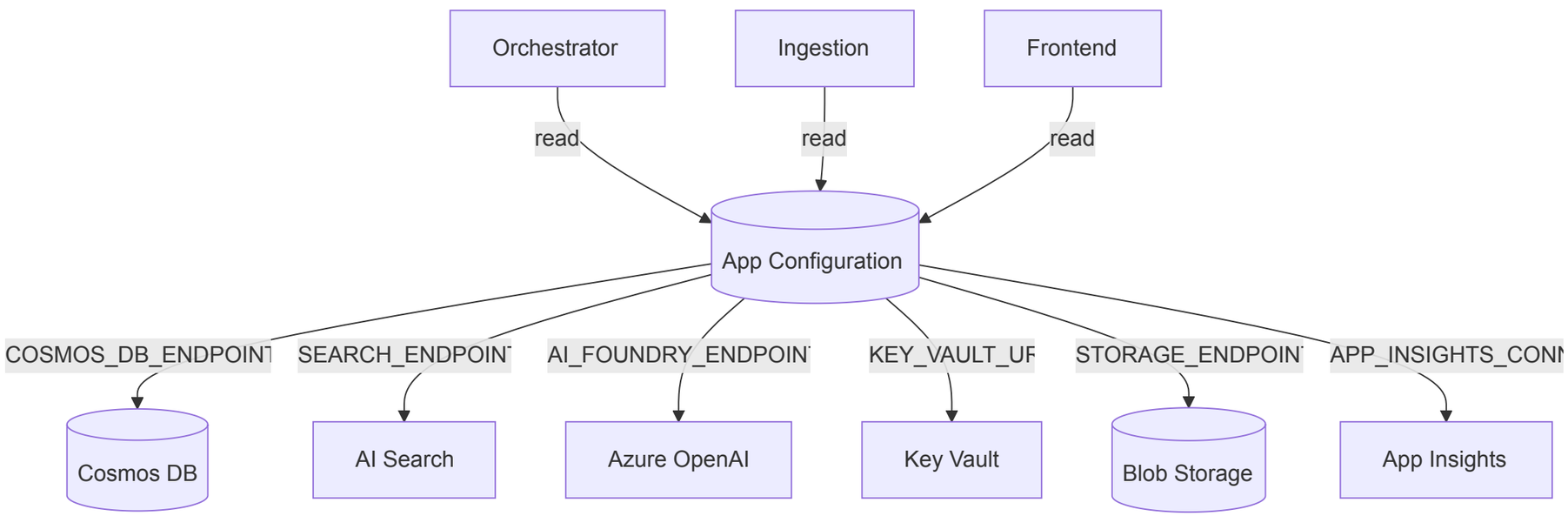
#	Category	Item	Status
1	Decision	Embedding model (recommend: large, 3072d)	
2	Decision	SharePoint sites/libraries to index	
3	Decision	Document-level security yes/no	
4	Decision	RAI blocked terms and severity thresholds	

#	Category	Item	Status
5	Decision	Azure region	⬜
6	Decision	AI Search tier (S1 for prod)	⬜
7	Azure	Request OpenAI quota (gpt-4o + embeddings)	⬜
8	Azure	Entra ID app registration + API permissions	⬜
9	Azure	Network mode (private endpoints vs public)	⬜
10	Config	Edit raiblocklist.json with org terms	⬜
11	Config	Edit raipolicies.json thresholds	⬜
12	Config	Verify search.j2 defaults	⬜
13	Config	Write custom system prompt for Cosmos DB prompts container	⬜

Deep Dive: Service-to-Service Connections

Every Azure service in GPT-RAG communicates through **Azure App Configuration** as the central service registry. At deployment time, Bicep populates App Configuration with endpoints, resource IDs, and connection strings for every resource. Each Container App reads App Configuration at startup to discover its dependencies — no hardcoded connection strings anywhere.

How Components Find Each Other



Per-Component RBAC Roles

Each Container App gets a **user-assigned managed identity** (UAI) with specific RBAC roles — no shared credentials, no secrets in environment variables.

Role	Orch	Ingest	Front	Purpose
AppConfigDataReader	✅	✅	✅	Read endpoints from App Config
CogSvcUser	✅	✅	—	AI Foundry services

Role	Orch	Ingest	Front	Purpose
CogSvcOpenAIUser	✅	✅	—	OpenAI models
CosmosDBContributor	✅	✅	—	Cosmos DB read/write
SearchIndexReader	✅	—	—	Query search index
SearchIndexContributor	—	✅	—	Write to search index
BlobDataReader	✅	—	✅	Read blobs
BlobDataContributor	—	✅	—	Write blobs
BlobDelegator	—	—	✅	SAS token generation
KeyVaultSecretsUser	✅	✅	✅	Read secrets
AcrPull	✅	✅	✅	Pull container images

Key insight: The orchestrator has **read-only** search access while ingestion has **write** access. Frontend has no AI or search access at all. This is **least privilege** in action.

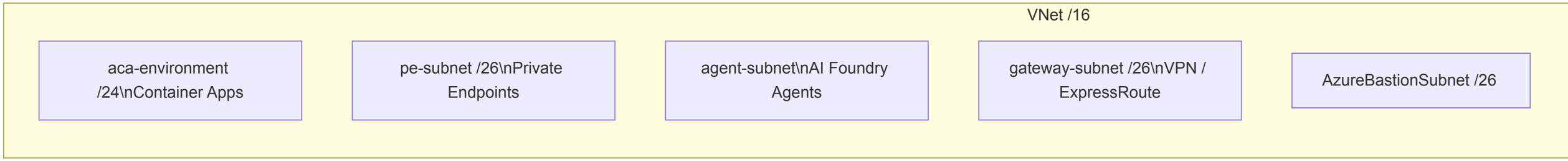
Communication Protocols

From	To	Auth
User browser	Frontend	Entra ID login
Frontend	Orchestrator	API key / managed identity
Orchestrator	OpenAI	Managed identity
Orchestrator	AI Search	Managed identity + user token
Orchestrator	Cosmos DB	Managed identity
Ingestion	SharePoint	Entra app registration
Ingestion	OpenAI	Managed identity
Ingestion	AI Search	Managed identity
Ingestion	Blob Storage	Managed identity
All apps	App Config	Managed identity
All apps	Key Vault	Managed identity
All apps	App Insights	Connection string

Deep Dive: VNet & Private Endpoint Topology

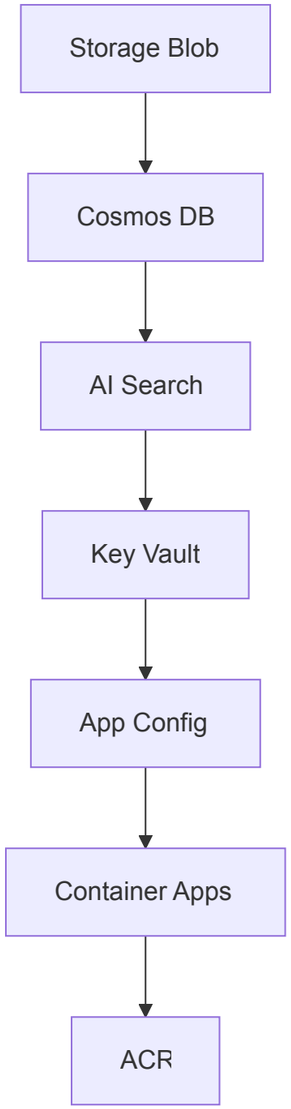
When `networkIsolation` is enabled (default for production), GPT-RAG creates a VNet with dedicated subnets and private endpoints for every Azure service. No service is publicly accessible.

VNet Subnet Layout



Private Endpoints (pe-subnet)

Serialized deployment to avoid race conditions:



Private DNS Zones

DNS Zone	Service
`privatelink.blob.core.windows.net`	Storage Blob
`privatelink.documents.azure.com`	Cosmos DB

DNS Zone	Service
`privatelink.search.windows.net`	AI Search
`privatelink.vaultcore.azure.net`	Key Vault
`privatelink.azconfig.io`	App Configuration
`privatelink.cognitiveservices.azure.com`	Cognitive Services
`privatelink.openai.azure.com`	Azure OpenAI
`privatelink.services.ai.azure.com`	AI Foundry
`privatelink.applicationinsights.io`	Application Insights
`privatelink.{region}.azurecontainerapps.io`	Container Apps
`privatelink.azurecr.io`	Container Registry

Each DNS zone is linked to the VNet. Container Apps resolve private IPs — no public traffic between components.

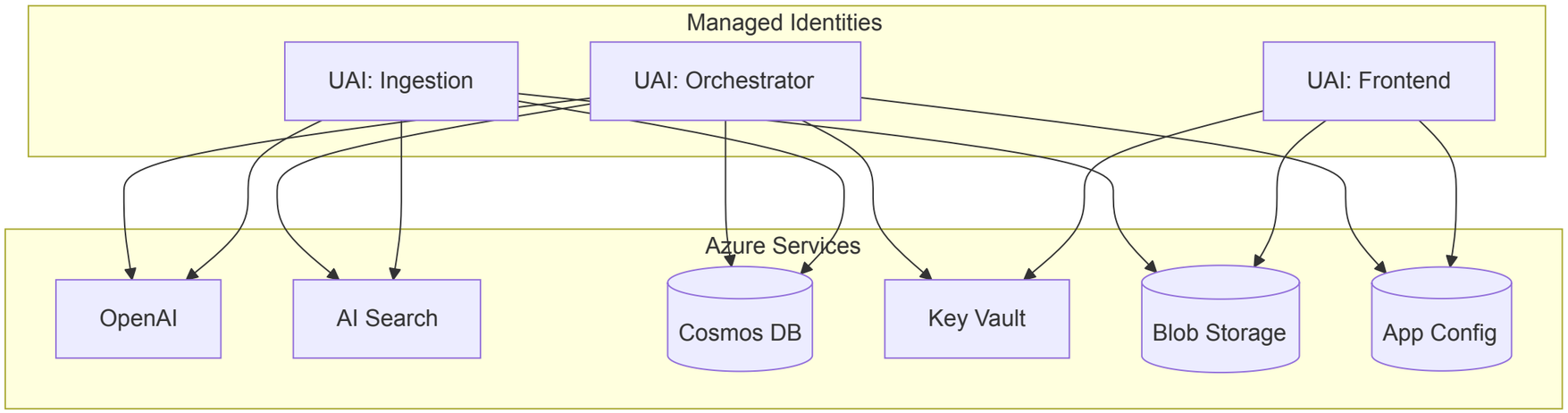
What This Means in Practice

- **Container Apps** run in `aca-environment-subnet` and reach all services through private DNS
- **All Azure services** sit behind private endpoints on `pe-subnet`
- **No public internet traffic** between components
- **Bastion + Jumpbox** for secure admin access
- **agent-subnet** reserved for AI Foundry Agent Service (future use)

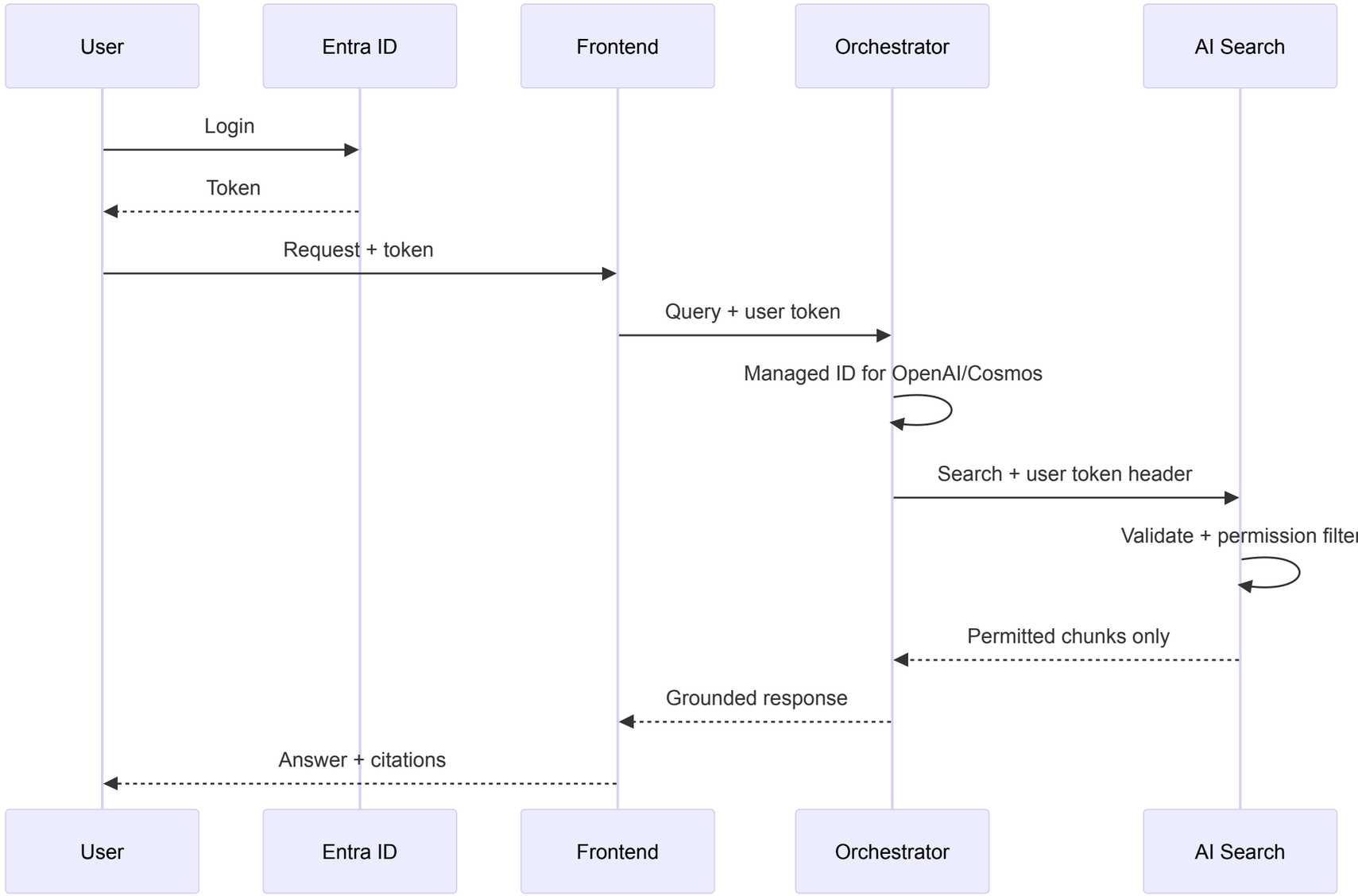
Deep Dive: Identity & Authentication Chain

GPT-RAG uses a **two-layer identity model**: user-assigned managed identities for service-to-service calls, and Entra ID tokens for end-user identity propagation.

Managed Identity Architecture



End-User Authentication Flow



Key Identity Concepts

Why two layers matter:

- **Managed identities** (service-to-service): The orchestrator uses its own UAI to call OpenAI, read/write Cosmos DB, and read Key Vault. No passwords, no API keys, no rotation needed. Bicep assigns RBAC roles automatically.
- **User token propagation** (document-level security): The end user's Entra ID token is forwarded from the frontend through the orchestrator to AI Search via the `x-ms-query-source-authorization` HTTP header. AI Search validates the token and applies permission filters (`groupIds` , `userIds` , `rbacScope`). This ensures users only see documents they have access to in SharePoint.

What happens without a user token: AI Search returns zero results (fail-closed design). This is intentional — no anonymous access to documents.

Deployer principal: During `azd provision` , the deploying user also receives roles like `CosmosDBBuiltInDataContributor` , `SearchServiceContributor` , `KeyVaultSecretsOfficer` so that the postprovision scripts can create indexes, inject secrets, and populate Cosmos DB containers.

App Configuration as Service Registry

Rather than each app discovering services independently, App Configuration acts as a central registry populated by Bicep at deployment time:

Config Key	Value	Used By
`COSMOS_DB_ENDPOINT`	Cosmos DB URI	Orch, Ingest
`SEARCH_SERVICE_QUERY_ENDPOINT`	AI Search endpoint	Orch, Ingest
`KEY_VAULT_URI`	Key Vault URL	All apps
`STORAGE_BLOB_ENDPOINT`	Blob endpoint	Ingest, Front
`AI_FOUNDRY_ACCOUNT_ENDPOINT`	OpenAI endpoint	Orch, Ingest
`MODEL_DEPLOYMENTS`	Model names + versions	Orch, Ingest
`CONTAINER_APPS`	App endpoints + FQDNs	Frontend
`AGENT_STRATEGY`	`single_agent_rag`	Orchestrator
`APPLICATIONINSIGHTS_CONNECTION_STRING`	App Insights	All apps
`NETWORK_ISOLATION`	`true` / `false`	All apps
`ENABLE_AGENTIC_RETRIEVAL`	`true` / `false`	Orchestrator

This means: if you need to change a model deployment or toggle agentic retrieval, you update App Configuration — no redeployment of Container Apps needed.